

BRAC UNIVERSITY
Department of Computer Science and Engineering
CSE422 : Artificial Intelligence
Assignment 02

- 1) You are tasked with designing a seating arrangement for a wedding reception with 8 guests. The guests have specific preferences about whom they would like to sit next to and whom they want to avoid. You need to create a seating arrangement for these guests around a circular table. Depending on the guests' preferred seating arrangements, the aim is to optimize their level of satisfaction. Consider the following constraints:
- a) Guest 1 (G1) prefers to sit next to Guest 2 and Guest 3 but wants to avoid Guest 4.
 - b) Guest 2 (G2) prefers to sit next to Guest 1 and Guest 4 but wants to avoid Guest 5.
 - c) Guest 3 (G3) prefers to sit next to Guest 1 and Guest 6 but wants to avoid Guest 7.
 - d) Guest 4 (G4) prefers to sit next to Guest 2 and Guest 5 but wants to avoid Guest 8.
 - e) Guest 5 (G5) prefers to sit next to Guest 4 and Guest 6 but wants to avoid Guest 1.
 - f) Guest 6 (G6) prefers to sit next to Guest 3 and Guest 5 but wants to avoid Guest 2.
 - g) Guest 7 (G7) prefers to sit next to Guest 8 but wants to avoid Guest 3.
 - h) Guest 8 (G8) prefers to sit next to Guest 7 but wants to avoid Guest 4.

Answer the following questions based on the scenario above.

- a. Encode the problem and create four parent chromosomes.
- b. Suggest a possible fitness function and choose the two fittest Chromosomes.
- c. Do you think the regular crossover (as in the n-queen problem) will work in this scenario? If not, briefly explain why.
- d. Suggest a possible case of mutation in this context.

2) You are given the following leaf nodes:

80	30	25	35	55	20	05	62	40	10	70	15	50	45	60	75
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- A. Construct a game tree using **alpha-beta** pruning. Showcase the values of the alpha, beta, and the final value of the nodes. Also, show the pruned branches.

[Branching factor is 2]

- B. Find the best choice (strategy) starting from the root node of your tree.
- C. Does ordering of nodes help alpha beta pruning to improve its performance?
Explain your answer.